FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2021

PAPER –1

TEST DATE: 18-01-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. Let acceleration on inclined part (while going up as well as while going down) be a. Length of inclined part be l_1 and that of flat part be l_2

Total time period is given as

$$T = \left[\sqrt{\frac{2\ell_1}{a}} + \frac{\ell_2}{\sqrt{2a\ell_1}} + \sqrt{\frac{2\ell_1}{a}}\right] \times 2$$

If l_1 and l_2 is made 4 times, the time period doubles.

2. C

Sol. Potential at centre $V_C = V_Q + V_{in} = K\frac{Q}{r_0} + 0 = \frac{KQ}{r_0}$

Potential at centre = potential at all the points of the shell

$$\therefore \qquad V_P = \frac{KQ}{r_0}$$

$$V_{in}^P + V_Q^P = \frac{KQ}{r_0} \therefore V_{in}^P = \frac{KQ}{r_0} - \frac{KQ}{r}$$

3. A

Sol. Consider an elemental length dx. Velocity of the element is perpendicular to the plane of the figure and its magnitude is

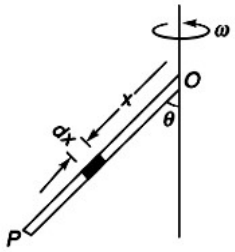
$v = \omega(x\sin\theta)$

Emf induced in the element is

$d\varepsilon = B(dx)\sin\theta$ [θ is angle between dx and B]

$= B\omega\sin^2\theta x dx$

$\therefore\ \varepsilon = B\omega\sin^2\theta\int_0^L x dx = \frac{1}{2}B\omega\sin^2\theta(L^2)$



The free electrons experience force towards P. Hence O is positive.

4. D

Sol. $\frac{dN}{dt} = t^2 - \lambda N$

For $\frac{dN}{dt}$ to be minimum $\frac{d^2N}{dt^2} = 0$

$\Rightarrow \quad \frac{d^2N}{dt^2} = 2t - \lambda(t^2 - \lambda N) = 0$

or, $\quad N = \frac{2t_0 - \lambda t_0^2}{\lambda^2}$

5. A

Sol. When a pressure difference $P_1 - P_2 = \Delta P$ is applied various height and length are as shown in the second figure.

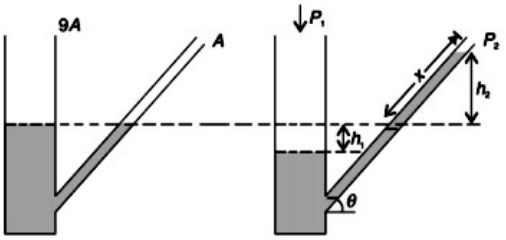
Since volume of manometer liquid is constant, we have

$\therefore 9Ah_1 = Ax \Rightarrow x = 9h_1$

$h_2 = x\sin\theta = 9h_1\sin\theta$

$\Delta P = \rho g(h_1 + h_2)$ [ρ = density of liquid]

$= 0.74\rho_w g\left[\frac{x}{9} + x\sin\theta\right] = 0.74\rho_\omega gx\left[\frac{1}{9} + \sin\theta\right]$



It is needed that when $\Delta P = 0.09 \times 10^{-3}\rho_\omega g$, value of x is 0.5×10^{-3} m

$\therefore 9\times10^{-5}\times\rho_\omega g = 0.74\rho_\omega g\times5\times10^{-1}\left[\frac{1}{9} + \sin\theta\right]$

$\frac{0.9}{0.74\times5} = \frac{1}{9} + \sin\theta$

$\sin\theta = 0.24 - 0.11 = 0.13$

$\therefore\ \theta = \sin^{-1}(0.13)$

6. B

Sol. In equilibrium the particle is at the lowest position. Consider the system at an angular position θ.

$\tau = mgR\sin\theta \simeq mgR\theta$ (for small θ)

$I\alpha = \tau$

$\left[\frac{1}{2}(2m)R^2 + mR^2\right]\alpha = -mgR\theta$

$\therefore \alpha = -\left(\frac{g}{2R}\theta\right) \therefore f = \frac{1}{2\pi}\sqrt{\frac{g}{2R}}$

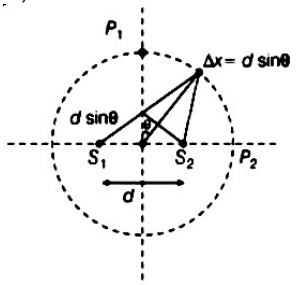
7. B, D

Sol. Values of options (B) and (D) do not match with the mirror formula.

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

8. C, D

Sol.



$\lambda = 600$ nm

At P_1, $\Delta x = 0$

At P_2, $\Delta x = 1.8$ mm $= n\lambda$

Number of maximas will be

$$n=\frac{\Delta x}{\lambda}=\frac{1.8\,mm}{600\,nm}=3000$$

At P_2, $\Delta x = 3000\lambda$

Hence, bright fringe will be formed

At P_2, 3000th maxima is formed

For option (a),

$\Delta x = d\sin\theta$

$d\Delta x = d\cos\theta d\theta$

$$R\lambda = d\cos\theta R d\theta$$

$$\Rightarrow \quad Rd\theta=\frac{R\lambda}{d\cos\theta}$$

As, we move from P_1 to P_2

$$\theta\uparrow \cos\theta\downarrow Rd\theta\uparrow$$

9. A, C, D

Sol. $$^{210}_{84}Po \rightarrow {}^{4}_{2}He + {}^{206}_{82}X$$

So, daughter nucleus is $^{206}_{82}X$

Disintegration energy is

$$Q=\frac{A}{A-4}\times KE_{\alpha}$$

$$=\left(\frac{210}{210-4}\right)\times 5.30=5.40\,MeV$$

Mass equivalence of disintegration energy is

$$M_Q=\frac{5.40}{931}=0.0058\,u$$

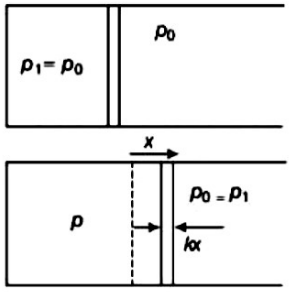
So, mass of daughter nucleus is

$$=m_{Po}-m_{\alpha}-m_Q$$

$$=209.9829-4.0026-0.0058=205.9745\,u$$

10. A, B

Sol. This question can be solved, if right hand side chamber is assumed open, so that its pressure remains constant even, if the piston shifts towards right.



$$pV = nRT$$

$$\Rightarrow \quad p \propto \frac{T}{V}$$

Temperature is made three times and volume is doubled,

$$p_2 = \frac{3}{2}p_1$$

Further,

$$x = \frac{\Delta V}{A} = \frac{V_2 - V_1}{A} = \frac{2V_1 - V_1}{A} = \frac{V_1}{A}$$

$$p_2 = \frac{3p_1}{2} = p_1 + \frac{kx}{A}$$

$$\Rightarrow \quad kx = \frac{p_1 A}{2}$$

Energy of spring,

$$\frac{1}{2}kx^2 = \frac{p_1 V_1}{4}$$

Also, $\Delta U = nC_V \Delta T$

$$= \frac{p_1 V_1}{RT_1} \times \frac{3}{2}R \times (3T_1 - T_1) = 3p_1 V_1$$

11. B, C

Sol. From the behaviour of electric lines, we can say that Q_1 is positive and Q_2 is negative

Further, $|Q_1| > |Q_2|$

At some finite distance to the right of Q_2, electric field will be zero. Because electric field due to Q_1 is towards right (away from Q_1) and due to Q_2 is towards left (towards Q_2). But since magnitude of Q_1 is more, the two fields may cancel each other, because distance of that point from Q_1 will also be more.

12. A, B, C, D

Sol. At low temperature (60 K), the hydrogen molecule has only translator motion, therefore it has only 3 degrees of freedom.

So, for 1 mole of gas,

$$U = \frac{3}{2}N_A k_B T = \frac{3}{2}RT$$

$$\Rightarrow \quad C_V = \frac{3}{2}R \text{ at } 60K$$

At temperatures above 60 K, i.e. between 250 K to 750 K, molecule of hydrogen began to rotate and behaves like a dumb-bell. It has now 5 degrees of freedom.

So, $U = \frac{5}{2}N_A k_B T = \frac{5}{2}RT$

or $C_V = \frac{5}{2}R$, at 250 to 750 K

At temperature around 4000 K, the molecular collisions are so vigorous that the atoms begins to vibrate and the molecules now have 7 degrees of freedom.

$\therefore \quad C_V = \frac{7}{2}R$ at 4000K

SECTION – C

13. 00006.00

Sol. $\frac{L_{total}}{L_B} = \frac{(I_A + I_B)w}{I_B.w} = \frac{m_A r_A^2}{m_B r_B^2} + 1 = \frac{11}{2.2} + 1 = 6$

14. 00004.20

Sol. When an excited atom emits a photon of momentum p, conservation of momentum requires that atom recoil with same momentum p.

For emitted photon,

$$\Delta E = \frac{hc}{\lambda} = pc$$

$\therefore \quad \frac{hc}{\lambda} = p$

Here, $\Delta E = E_5 - E_1$

$$= \left(-\frac{13.6}{25}\right) - \left(\frac{-13.6}{1}\right)$$

$$= 13.6 \times \frac{24}{25} \times 1.6 \times 10^{-19} J$$

Momentum of photon,

$$p = \frac{13.6 \times 24 \times 1.6 \times 10^{-19}}{25 \times 3 \times 10^8}$$

$$= 6.96 \times 10^{-27} kg\, ms^{-1}$$

So, recoil speed of H-atom is

$$\frac{p}{m} = \frac{7 \times 10^{-27}}{1.67 \times 10^{-27}} = 4.20\, ms^{-1}$$

15. 00000.95

Sol. $U = \frac{1}{2}\left(\frac{Mg}{\pi r^2}\right)\left(\frac{\Delta l}{l}\right)(\pi r^2 l) = 0.95\ J$

16. 00075.00

Sol. Work done at constant pressure is

$$\Delta W = \left(1 - \frac{1}{r}\right) \times \Delta Q_{total}$$

$\Rightarrow \quad 25 = \left(1 - \frac{1}{r}\right)\Delta Q_{total}$

But, $r = 1 + \frac{2}{f} = 1 + \frac{2}{6}$

or $r = \frac{4}{3}$

So, $25 = \left(1 - \frac{3}{4}\right) \times \Delta Q_{total}$

or $\Delta Q_{total} = 100\ J$

So, net heat absorbed by the gas = 100 – 25 = 75 J

17. 00000.60

Sol. $\left|\frac{v}{u}\right| = 2$ or $|u| = 2|u|$

When u = –x, then v = + 2x and $|x| + |v| = 1.8m$

$\Rightarrow$ $3x = 1.8m$ or $x = 0.60\ m$

18. 00000.40

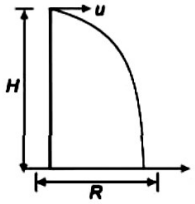
Sol. $H = \frac{1}{2}gt^2$

$\Rightarrow$ $500 = \frac{1}{2} \times 10 \times t^2$

$\Rightarrow$ t = 10s

Also, R = ut

$\Rightarrow$ $u = \frac{R}{t} = 40\ ms^{-2}$



If recoil velocity of gun is v, then

$m_1 v = m_2 u$

or $v = 100 \times 40 = 0.4\ ms^{-1}$

Chemistry

PART – II

SECTION – A

19. B

Sol. $80 \times 20 = 32 \times n$

$\Rightarrow n = 50$

20. B

Sol. $P^2V = \text{constant}$

$P \propto \frac{T}{V}$

$\Rightarrow \frac{T^2}{V^2} \times V = \text{constant}$

$\Rightarrow \frac{T^2}{V} = \text{constant}$

21. C

Sol. $H^-(aq) + OH^-(aq) \rightleftharpoons H_2O(aq)$

ΔH_f of $H^+(aq) = 0$

22. C

Sol. $[NH_2^-] = 10^{-15}$

$= 10^{-15}$ mol/L

$=6 \times 10^{23} \times 10^{-15}$ ions per mL

$\Rightarrow 6 \times 10^{23} \times 10^{-15} \times 10^{-3}$ ions per mL

23. D

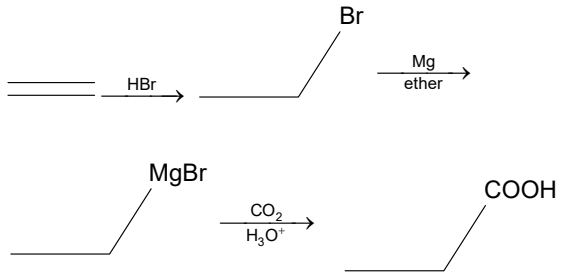
Sol. Factual

24. A

Sol. (F^{-1} is a weak field ligand)

25. B, C

Sol.



26. A, B

Sol. Factual

27. A, C, D

Sol. Factual

28. A, B, C, D
Sol. Factual (from structure)

29. B, C
Sol. Factual

30. A, B
Sol. Factual

SECTION – C

31. 00182.00
Sol. $2(r_{Li} + r_{Cl}) = \sqrt{2} \times 520$
and $2r_{Cl} = 520$

32. 00377.00
Sol. X = molarity of $MgSO_4$ = $\frac{12}{100} \times \frac{1000}{M_{MgSO_4}}$

Y = Molarity of $MgCl_2$ = $\frac{9.5}{100} \times \frac{1000}{M_{MgCl_2}}$

Total molarity of ions
$= 1.8x + 2.2y$
delta $T_b = K_d(1.8x + 2.2y)$

33. 00375.00
Sol. Reaction is first order w.r.t. B, so half life is constant if B is changed
Reaction is second order w.r.t. A so half life becomes half on double conc. of A
312.5 + 62.5 = 375

34. 00021.60
Sol. The specific rotation of (+2)-2-bromooctane is

$$[\alpha] = \frac{2.7}{\frac{5g}{40mL} \times 1dm} = 21.6^o$$

35. 00040.00
Sol. Ratio of $3^o : 1^o$ chloride
$= 6 \times 1 : 9 \times 1$
% of $3^o = \frac{6 \times 100}{6 + 9} = 40\%$

36. 00004.00
Sol. (b), (c), (e) and (h)

Mathematics

PART – III

SECTION – A

37. B

Sol. $\sum_{r=1}^{10} {}^{15}C_r.{}^{35}C_{10-r} = {}^{15}C_1.{}^{35}C_9 + {}^{15}C_2.{}^{35}C_8 + + + {}^{15}C_{10}.{}^{35}C_0$

$= {}^{50}C_{10} - {}^{35}C_{10}$

38. B

Sol. $f(x) = \int x^{\sin x}\left(1 + x.\cos x.\ln x + \sin x\right)dx$

If $f(x) = x^{\sin x} = e^{\sin x.\ln x}$, then

$f(x) = \int\left(f(x) + xf'(x)\right)dx = x.x^{\sin x} + c$

$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2}.\left(\frac{\pi}{2}\right) + c \qquad \Rightarrow c = 0$

$f(x) = (x)(x)^{\sin x}$, $f(\pi) = \pi$

39. C

Sol. Let $\vec{a} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$

Then by definition, a vector orthogonal to $\vec{a}$ and coplanar to $\vec{b}$ and $\vec{c}$ is $\vec{a} \times \left(\vec{b} \times \vec{c}\right)$

$\therefore$ Required unit vector s $= \frac{\vec{a} \times (\vec{b} \times \vec{c})}{\left|\vec{a} \times (\vec{b} \times \vec{c})\right|} = \frac{3\hat{j} - \hat{k}}{10}$.

40. C

Sol. $\sum_{r=1}^{n} \frac{2r+1}{r^2(r+1)^2} = \sum_{r=1}^{n}\left(\frac{1}{r^2} - \frac{1}{(r+1)^2}\right)$

$S_{10} = \left(\frac{1}{12} - \frac{1}{2^2}\right) + \left(\frac{1}{2^2} - \frac{1}{3^2}\right) + ... + \left(\frac{1}{10^2} - \frac{1}{11^2}\right)$

$= 1 - \frac{1}{121} = \frac{120}{121}$.

41. B

Sol. $\cos^{-1}\left(2x^2 - 1\right) = 2\pi - 2\cos^{-1} x \ \ (x < 0)$

$\therefore \cos^{-1}\left(2x^2 - 1\right) - 2\sin^{-1} x = 2\pi - 2\cos^{-1} x - 2\sin^{-1} x$

$= 2\pi - \pi = \pi$

42. C

Sol. $\begin{vmatrix} 1 & \cos(B-A) & \cos(C-A) \\ \cos(A-B) & 1 & \cos(C-B) \\ \cos(A-C) & \cos(B-C) & 1 \end{vmatrix} = \begin{vmatrix} \cos A & \sin A & 0 \\ \cos B & \sin B & 0 \\ \cos C & \sin C & 0 \end{vmatrix} \times \begin{vmatrix} \cos A & \sin A & 0 \\ \cos B & \sin B & 0 \\ \cos C & \sin C & 0 \end{vmatrix} = 0$

43. A, C, D

Sol. $az^4 + bz^3 + cz^2 + dz + e = 0$ (a, b, c, d $\in$ R)

Let roots of z_1, z_2, z_3, z_4

$|z_1| = |z_2| = |z_3| = |z_4| = 1$

$z_1 + z_2 + z_3 + z_4 = -a \quad \therefore |a| \le 4$

$z_1z_2 + z_2z_3 + z_1z_3 + z_1z_4 + z_2z_4 + z_3z_4 = b, \qquad (b) \le 6$

$z_1z_2z_3 + z_1z_2z_4 + z_1z_3z_4 + z_2z_3z_4 = -c, \qquad (c) \le 4$

44. B, C, D

Sol. $\lim\limits_{x\to\infty}\left(\dfrac{ax+1}{bx+2}\right)^x = \lim\limits_{x\to\infty}\left(\dfrac{a+\frac{1}{x}}{b+\frac{2}{x}}\right)^x = \left(\dfrac{a}{b}\right)^x$

If $0 < \dfrac{a}{b} < 1$, then $\left(\dfrac{a}{b}\right)^\infty = 0$

$\therefore \lim\limits_{x\to\infty} f(x) = 0$ for $0 < a < b$

If $\dfrac{a}{b} > 1$ then $\left(\dfrac{a}{b}\right)^\infty = \infty$

$\therefore \lim\limits_{x\to\infty} f(x)$ is non-existent for a > b > 0

If a = b, $\lim\limits_{x\to\infty} f(x) = e^{\lim\limits_{x\to\infty}\frac{-x}{ax+2}}$ or $e^{\lim\limits_{x\to\infty}\frac{-1}{a+\frac{2}{x}}}$

$= e^{-\frac{1}{a}}$ or $e^{-\frac{1}{b}}$

45. B, C, D

Sol. Sides are 14, 12, 10, Area of $\Delta ABC = \sqrt{18.4.6.16}$

$= 24\sqrt{6}$

$\Rightarrow$ sides of $\Delta$are in AP

$\therefore$ ex-radii are in HP

$r = \dfrac{\Delta}{S} = \dfrac{24\sqrt{6}}{18} = \dfrac{4\sqrt{6}}{3}$

46. A, D

Sol. $f(x) = \left(\sin^2 x - 1\right)^n$, $f\left(\dfrac{\pi}{2}\right) = 0$

$f\left(\dfrac{\pi^+}{2}\right) = \left(0^-\right)^n$, $f\left(\dfrac{\pi^-}{2}\right) = \left(0^-\right)^n$

If n is even, $f\left(\dfrac{\pi^+}{2}\right)$ and $f\left(\dfrac{\pi^-}{2}\right) > 0$

Then $x = \dfrac{\pi}{2}$ is point of minima

If n is odd, $f\left(\frac{\pi^+}{2}\right)$ and $f\left(\frac{\pi^-}{2}\right) < 0$, then $x = \frac{\pi}{2}$ is point of maxima

47. B, C

Sol. Let $G(x) = e^x.\cos x.f(x)$

$G'(x) = e^x.\cos x.f'(x) + e^x(\cos x - \sin x)f(x)$

$= e^x\left(\cos x.f'(x) + (\cos x - \sin x)f(x)\right) \le 0$

As given that $\cos x f'(x) \le (\sin x - \cos x)f(x)$

G(x) is non-increasing function

$\therefore G(6) = e^6 \cos 6 f(6) \le 0$

$\therefore f(6) \le 0$

But it is given that f(x) is non-negative

$\therefore$ f(6) = 0, Similarly f(5) = 0, f(7) = 0

Thus $x^2 - 3x + 2 + f(7) = 0$ has two distinct solutions

$$\lim_{x\to 6}\frac{f(x) - \sin(\pi x)}{x-6} = \lim_{x\to 6}\frac{f'(x) - \pi\sin(\pi x)}{1} = f'(6) - \pi < 0$$

48. A, B, C, D

Sol. Line and line = ${}^8C_2(1) = 28$

Circle and circle = ${}^7C_2(2) = 42$

Triangle and triangle = ${}^6C_2.6 = 90$

Circle and circle + triangle and triangle

Circle and triangle = 42 + 90 + ${}^6C_1.{}^7C_1.6 = 384$

SECTION – C

49. 00000.00

Sol. $x^2 - (2y + k)x + 2y^2 - 2ky - 2 \ge 0 \ \forall x \in R$

$D \le 0$

$(2y + k)^2 - 4(2y^2 - 2ky - 2) \le 0$

$\Rightarrow 4y^2 + 4ky + k^2 - 8y^2 + 2ky + 8 \le 0$

$\Rightarrow -4y^2 + 12kxy + k^2 + 8 \le 0 \ \forall y \in R$

$\Rightarrow 4y^2 - 12kxy - (k^2 + 8) \ge 0 \forall y \in R$

$\Rightarrow D \le 0 \Rightarrow 144k^2 + 16(k^2 + 8) \le 0$

$k \in \phi$

50. 00007.00

Sol. $y = Ax^m + B^{-n} \ \therefore \frac{dy}{dx} = A.m.x^{m-1} - n.Bx^{n-1}$

$\frac{d^2y}{dx^2} = Am(m-1)x^{m-2} + n(n+1)Bx^{n-2}$

Put these values in $x^2.\frac{d^2y}{dx^2}+2x\frac{dy}{dx}=12y$

$m=3,\ -4,\ n=4,-3$

$\therefore$ max value of (m + n) = 7

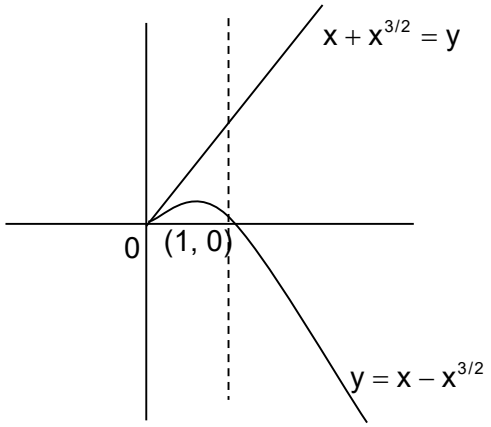
51. 00000.80

Sol. $(y-x)^2=x^3,\ x\geq 0$

$\Rightarrow y=x\pm x^{3/2}$

Required area

$=\int_0^1\left(\left(x+x^{3/2}\right)-\left(x-x^{3/2}\right)\right)dx=\frac{4}{5}\text{sq.unit}$



52. 00000.50

Sol. $I=\int_0^1\frac{\tan^{-1}x}{\tan^{-1}\left(\frac{(1-x)+x}{1-(1-x)x}\right)}dx=\int_0^1\frac{\tan^{-1}x}{\tan^{-1}(1-x)+\tan^{-1}x}$

$I=\int_0^1\frac{\tan^{-1}(1-x)}{\tan^{-1}(x)+\tan^{-1}(1-x)}dx \qquad 2I=1$

$I=\frac{1}{2}=0.5$

53. 00001.00

Sol. Let parabola be $y^2=4ax$ and $P\left(at_1^2,\ 2at_2\right)$ and $Q\left(at_2^2,2at_2\right)$, T be the point of intersection of tangent at t_1 and t_2

$= T\left(at_1t_2,\ a\left(t_1+t_2\right)\right)$

$P'\equiv\left(at_3t_1,\ a\left(t_3+t_1\right)\right)$

$Q'\equiv\left(at_2t_3,a\left(t_2+t_3\right)\right)$

$\frac{TP'}{TP}=\frac{\lambda}{1}\Rightarrow\lambda=\frac{t_3-t_2}{t_1-t_2}$

Similarly $\frac{TQ'}{TQ}=\frac{t_1-t_3}{t_1-t_2}$

$\therefore\frac{TP'}{TP}+\frac{TQ'}{TQ}=1$

54. 00080.00

Sol. Let sides of rectangle be p and q

$pq = 200$

Let an of ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with eccentricity e

$\pi ab = 200\pi \quad \Rightarrow ab = 200$

$BD = \sqrt{p^2+q^2}$

$p^2+q^2 = 4a^2e^2$ (distance between foci = 2ae)

$(p+q)^2 - 2qp = 4\left(a^2b^2\right)$ ...(1)

Since sum of distance of any point an ellipse from both foci is equal to length of major axis

$\therefore$ p + q = 29

Equation (1) becomes

$4a^2 - 2pq = 4a^2 - 4b^2, \therefore 4b^2 = 400$

$b = 10$